Algebra 1 Name___________________
Unit 10 Date___________________
Lesson 2 Homework Period___________________

Use the following context to answer questions 1-5.

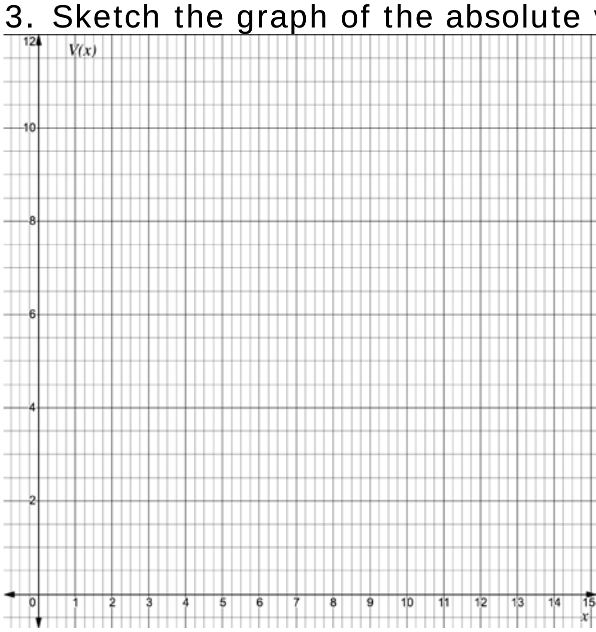
The average mass of a baseball is $8\frac{7}{8}$ ounces (oz) with a tolerance of $3\frac{7}{8}$ oz. If x is the mass of the baseball and V is the variation from $8\frac{7}{8}$ oz., the function $V(x) = \left|x - 8\frac{7}{8}\right|$ can be used to find the amount of variation.

1. Determine the maximum and minimum x-values.

2. Complete the table for $V(x) = \left|x - 8\frac{7}{8}\right|$. Use the minimum x-value in the first column, the maximum x-value in the last column, and add $\frac{31}{16}$ to each x-value in between.

x					
y					

3. Sketch the graph of the absolute value function.



4. Interpret the vertex of $V(x) = \left|x - 8\frac{7}{8}\right|$.

5. Interpret the axis of symmetry of $V(x) = \left|x - 8\frac{7}{8}\right|$.

Use the following context to answer questions 6-10.

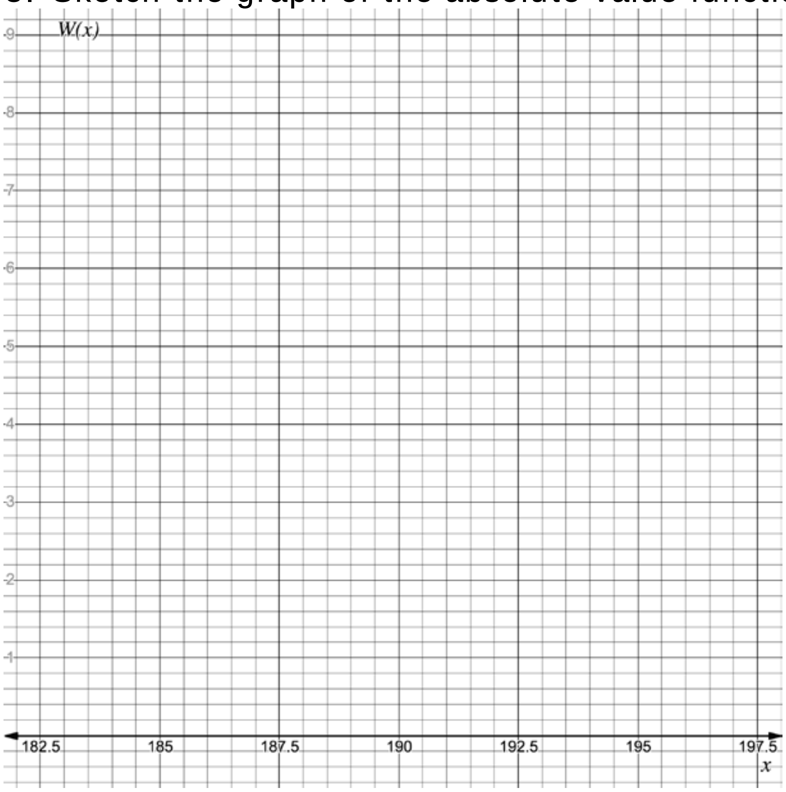
The clearance below a bridge is 190.5 feet (ft) above sea level. According to the distribution of tidal phases, the water under the bridge can fluctuate up to 4.5 feet above or below the normal depth. If x is the amount of clearance the bridge has above the water, in ft, and W is the variation in water depth away from sea level, in ft, the equation to find the amount of variation is $V(x) = |x - 190.5|$.

6. Determine the maximum and minimum clearance between the bridge and water.

7. Complete the table for $V(x) = |x - 190.5|$.

x	186	187.5	189	190.5	192	193.5	195
y							

8. Sketch the graph of the absolute value function.



9. Interpret the vertex of $V(x) = |x - 190.5|$.

10. Interpret the axis of symmetry of $V(x) = |x - 190.5|$.